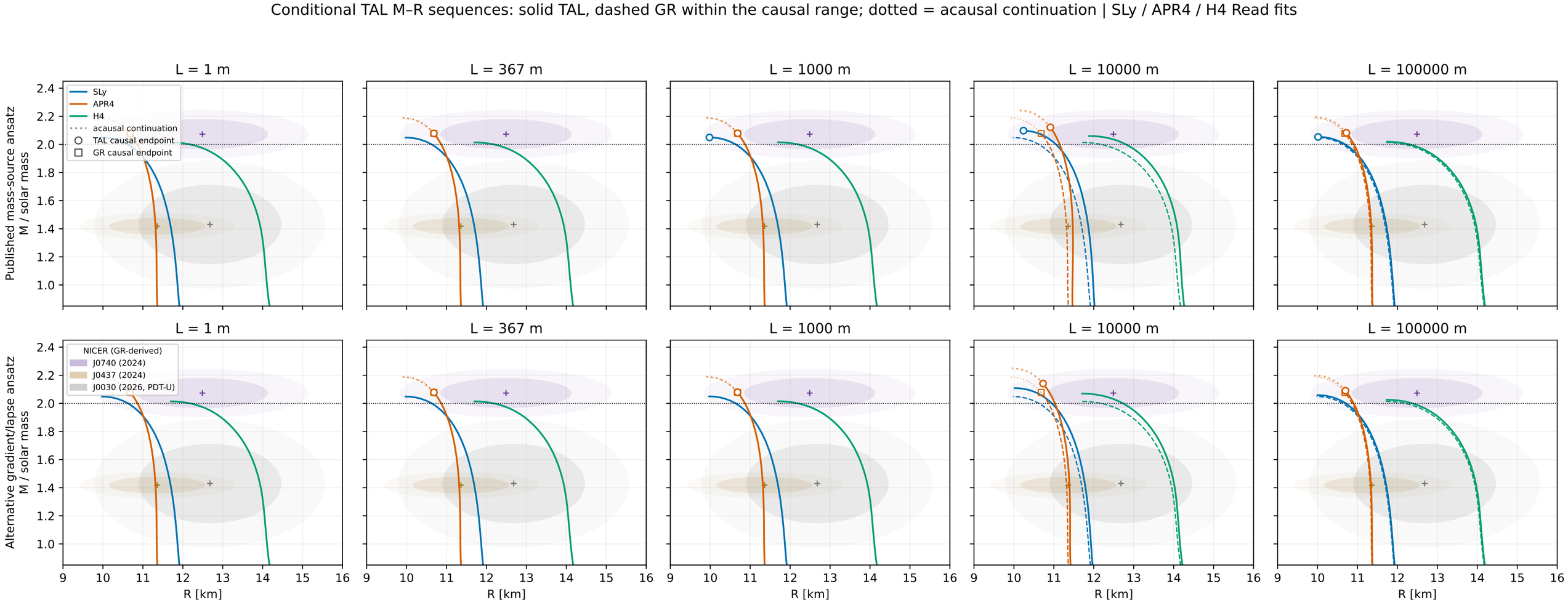

Conditional TAL M–R sequences: solid TAL, dashed GR within the causal range; dotted = acausal continuation | SLy / APR4 / H4 Read fits
L = 1 m
L = 367 m
L = 1000 m
L = 10000 m
L = 100000 m
SLy
APR4
H4
acausal continuation
TAL causal endpoint
GR causal endpoint
NICER (GR-derived)
J0740 (2024)
J0437 (2024)
J0030 (2026, PDT-U)
Published mass-source ansatz
M / solar mass
Alternative gradient/lapse ansatz
M / solar mass
R [km]
Shading: independent, symmetrized Gaussian 68% / 95% JOINT approximations (not published NICER contours). Purple: J0740 (2024); brown: J0437 (2024); gray: J0030 (2026, PDT-U).
Nonrotating TOV; GR-derived NICER posteriors are illustrative references, not a TAL likelihood. Modified-branch stability is not established.
Curves are solid/dashed up to the causal endpoint (c_s^2=c^2 first reached at the centre; open circle TAL, open square GR); the dotted, faded continuation up to the sequence maximum
is a potentially acausal benchmark of the published fit (SLy: 3 of 11 sequences; APR4: 11 of 11 sequences). No causal completion was imposed.